REPURPOSING COLLABORATION OPPORTUNITY

VON WILLEBRAND FACTOR HUMAN -> Costello syndrome

For clinical-collaborator outreach.

Why this hypothesis is real

Drug-target convergence on the disease scores **0.79** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Lead target is well expressed in the disease-relevant tissue (heart left ventricle). Reactome co-membership reinforces the indirect mechanism with **11** shared specific pathways.

Why now (IP runway + literature footprint)

The substance is a biologic or otherwise outside the FDA Orange Book; check the Purple Book and direct registry for IP runway.

Clinical opportunity

US population: approximately **166** patients -- below the 200,000 FDA Orphan Drug Act threshold. Orphan exclusivity + premium pricing changes the deal math materially. (source: Orphanet Point prevalence (<1 / 1 000 000, Worldwide))

What we propose

BioBix (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:

- Target validation across blood / immune / cancer multi-omics datasets
- Biomarker candidate identification from public + private cohorts
- Patient stratification analytics for trial-enrichment design
- Mechanism-of-action mining via systems-biology priors
- Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

- Investigator-initiated trial framework with the substance owner
- BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
- Joint publication strategy; data-sharing agreement gated on IP terms

Next step

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

Wim Van Criekinge, Professor (PhD, h-index 73)
Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University
`Wim.VanCriekinge@UGent.be`